

4 (long)

(a) Draw the small signal model for the circuit shown in Fig. 4 and derive the expressions for:

- (i) Input impedance
- (ii) No-load gain
- (iii) Output impedance

Determine the values of the above circuit parameters if $g_m = 5 \text{ mS}$, $r_d = 20 \text{ k}\Omega$, $R_1 = 10 \text{ M}\Omega$ and $R_2 = 20 \text{ k}\Omega$. [16]

(b) What is the function of this circuit? Suggest an application and briefly explain why this circuit would be useful. [6]

(c) If the circuit drives an inductive load of 25 mH , calculate the frequency at which the load current drops to 70% of its mid-band value. [8]

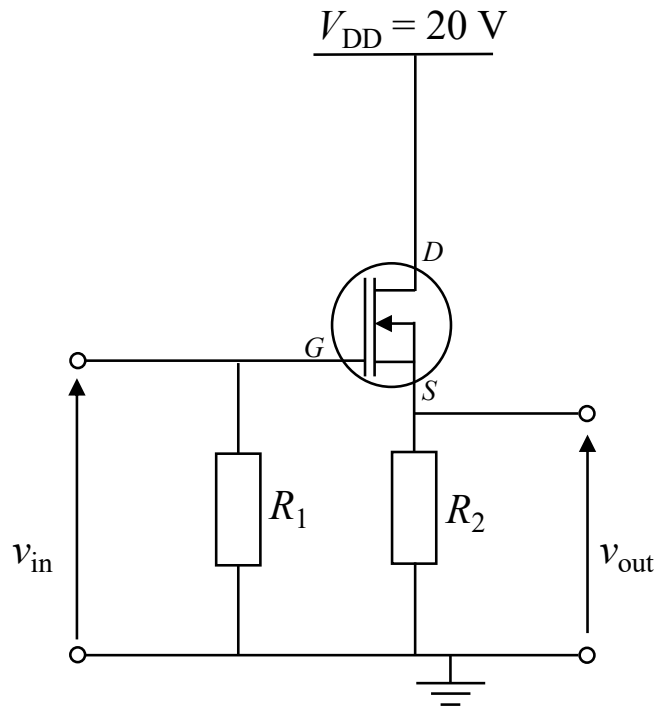


Fig. 4